

TAPE: Tool-Guided Adaptive Planning and Constrained Execution in Language Model Agents

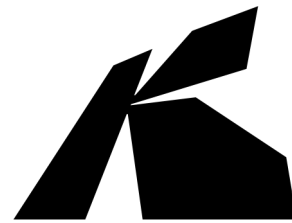
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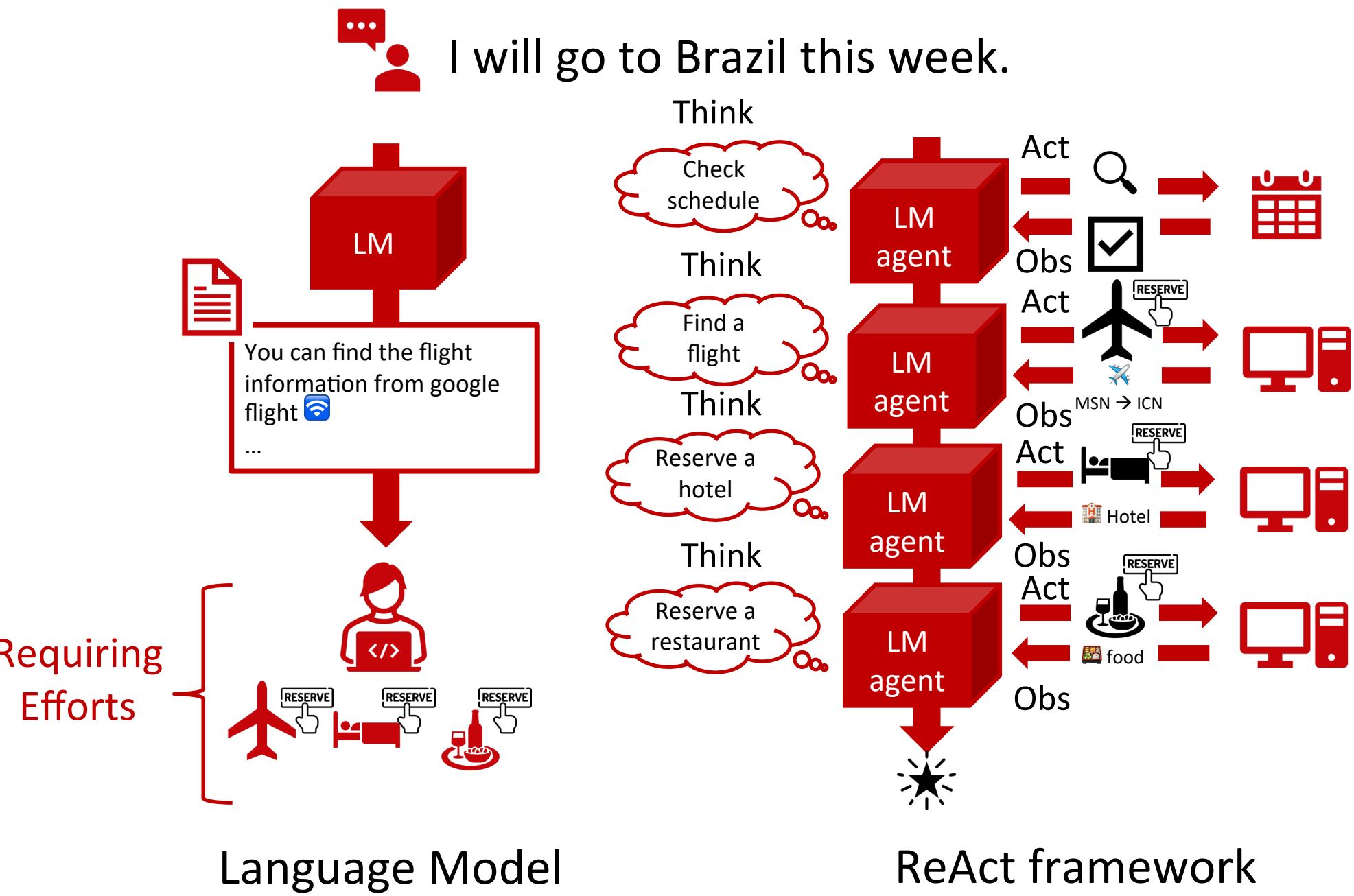


ludo robotics

How can Language Model (LM) agents maximize the success rate on tasks in the presence of irrecoverable failures?

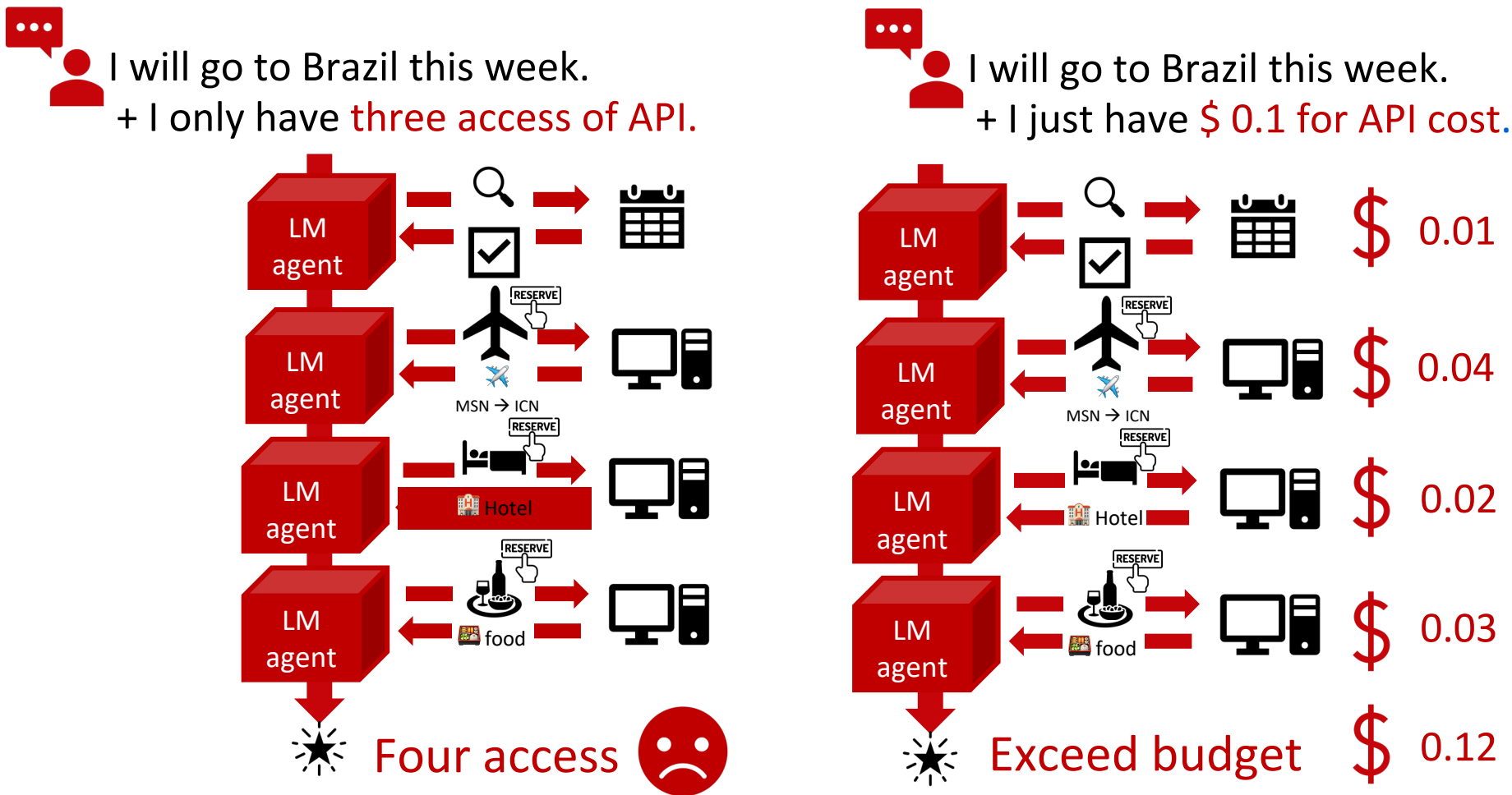
Language Model (LM) agent

- LM agents, especially ReAct-based agents, extend the LM's capability from generating text **to solving the task itself** by **iteratively think, act via tools, observe** from the external environment.



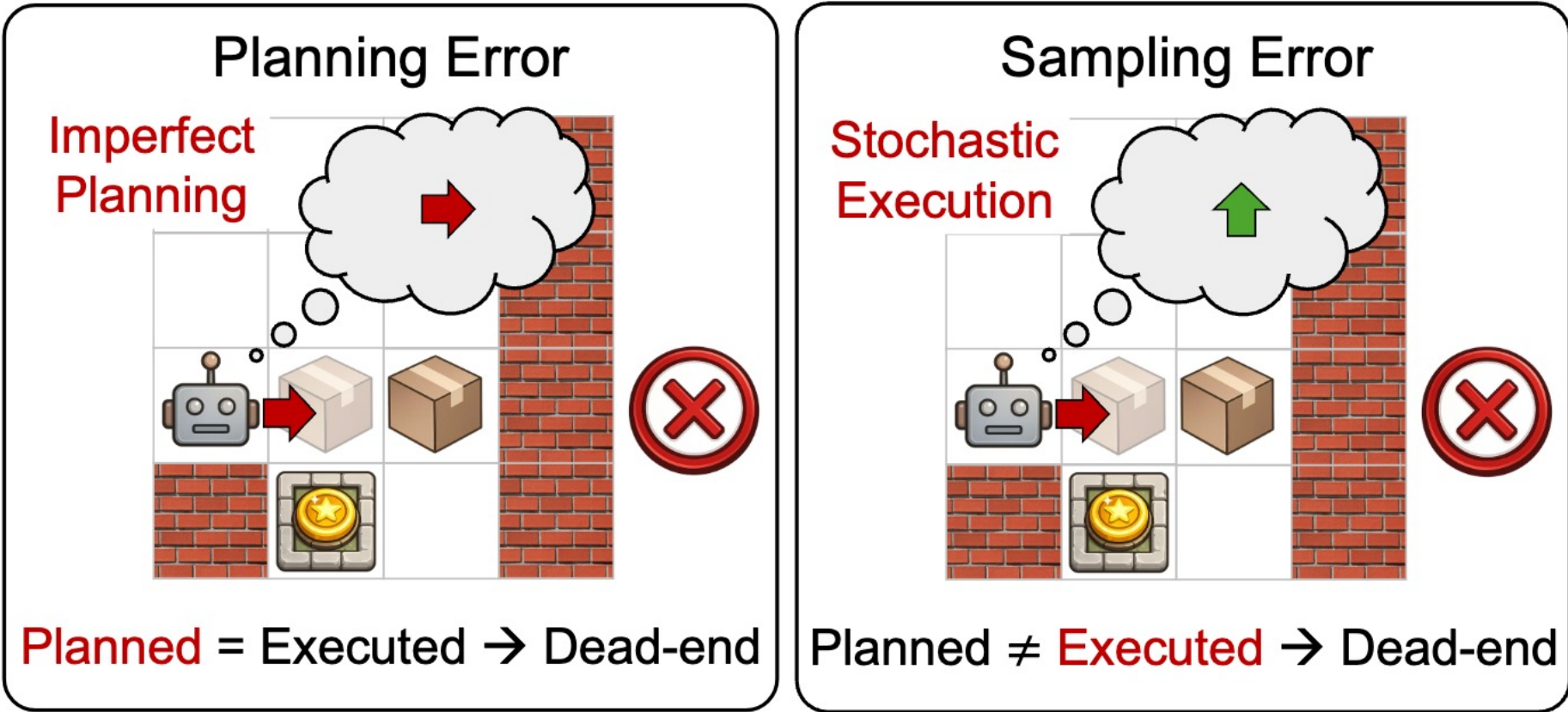
Real-world agent tasks are difficult

- Agents cannot rely on unlimited resources.
- Agent tasks involve constraints, such as # tool calls, total budget, and deadlines, **making the task difficult**.

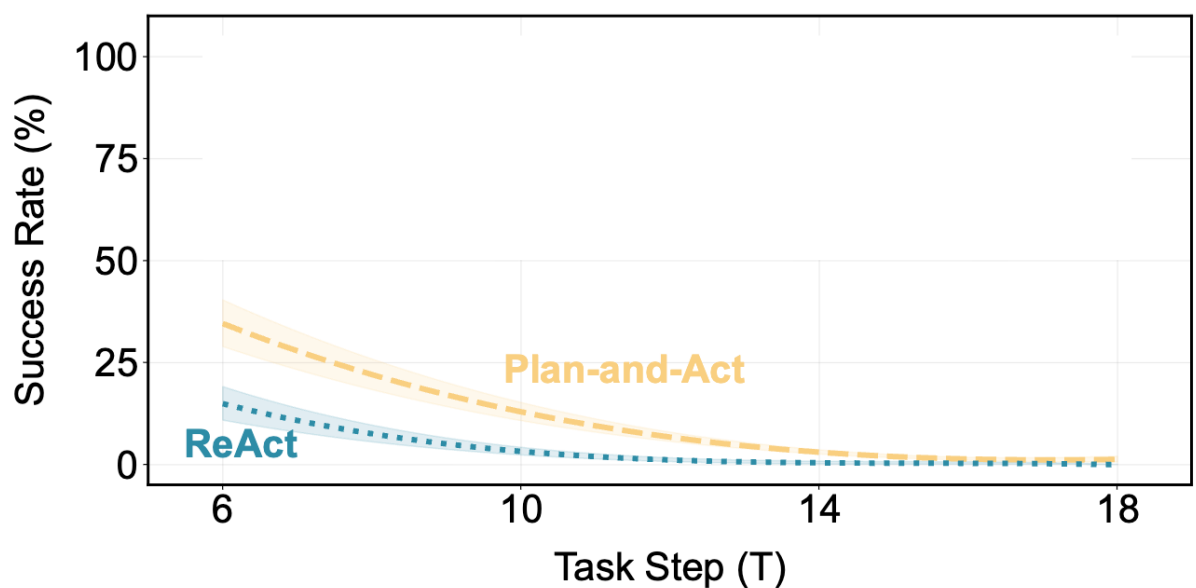


Two Sources of Irrecoverable Failures

- Planning Error:** Reasoning suggests an action towards irrecoverable failures.
- Sampling Error:** Token sampling deviates from the plan.

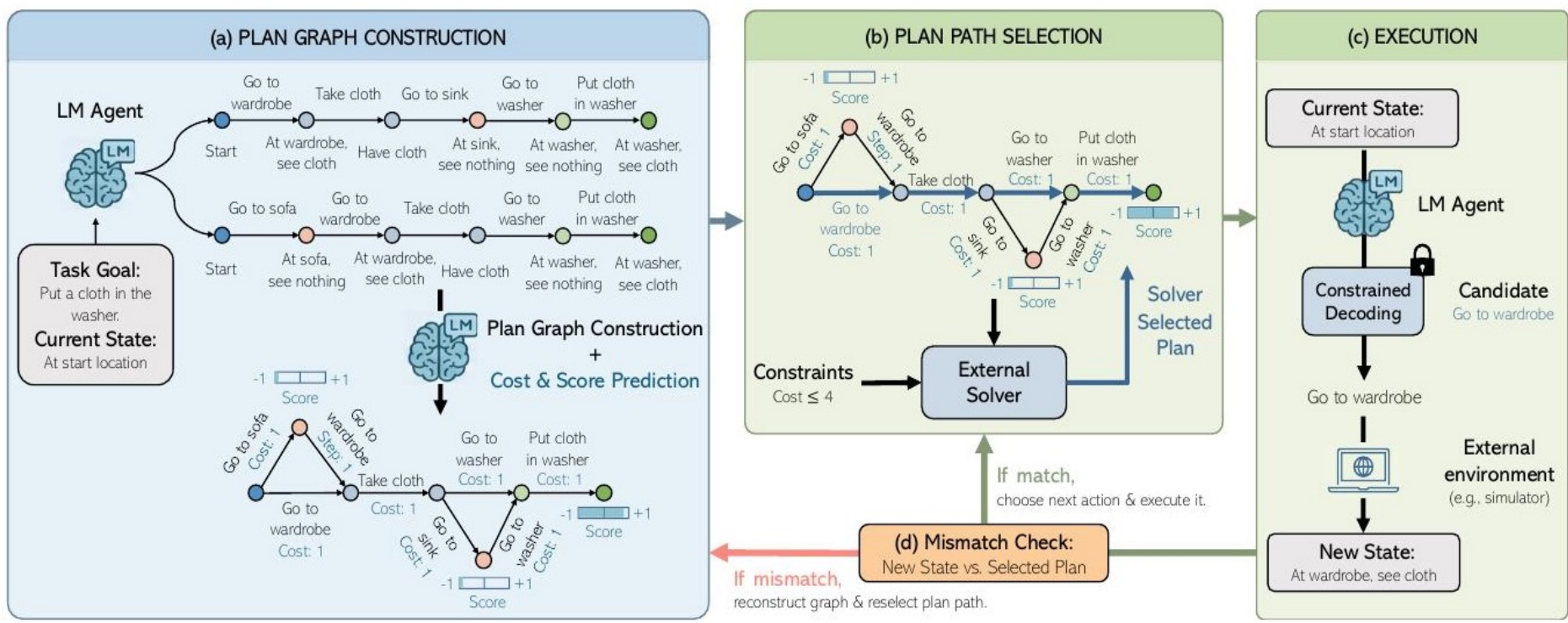


- Our toy example:** Error compounds
→ reducing success rate

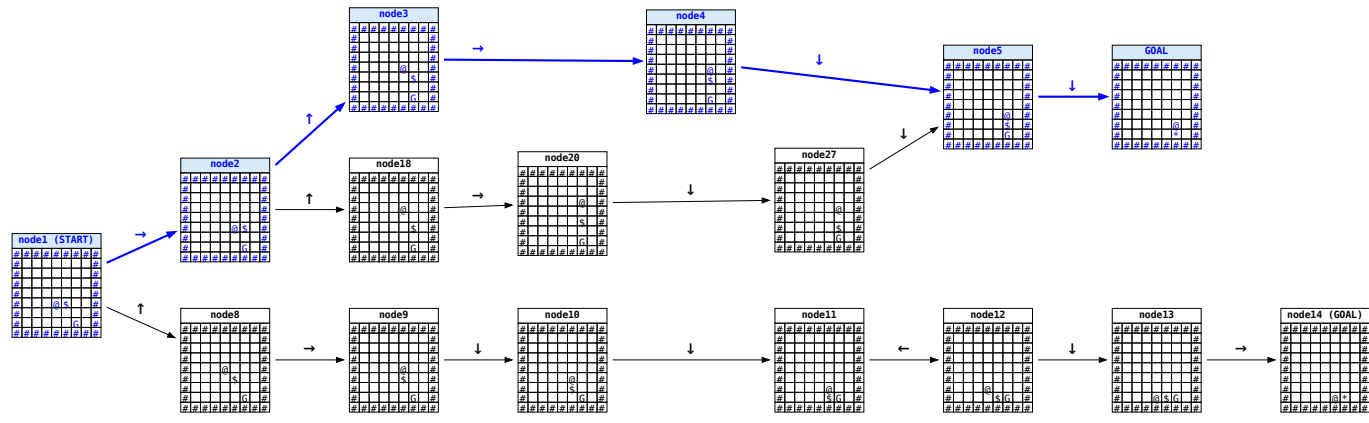


Our method: TAPE

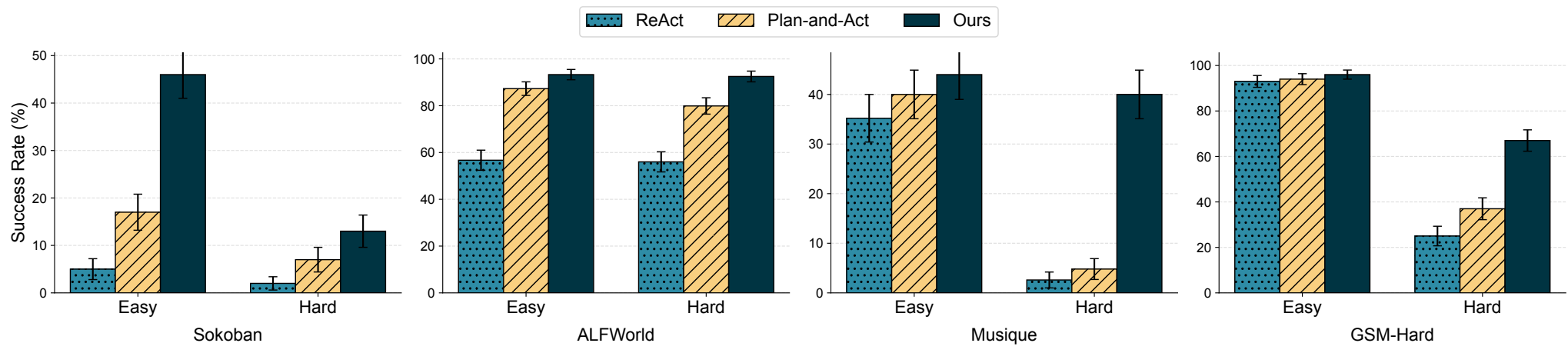
- TAPE samples **multiple plans** and aggregates them into a **plan graph**.
- An **external solver (e.g., ILP)** picks a feasible path.
- The agent executes via **constrained decoding**.
- TAPE **replans** where mismatch is detected.



- Graph & A selected path example

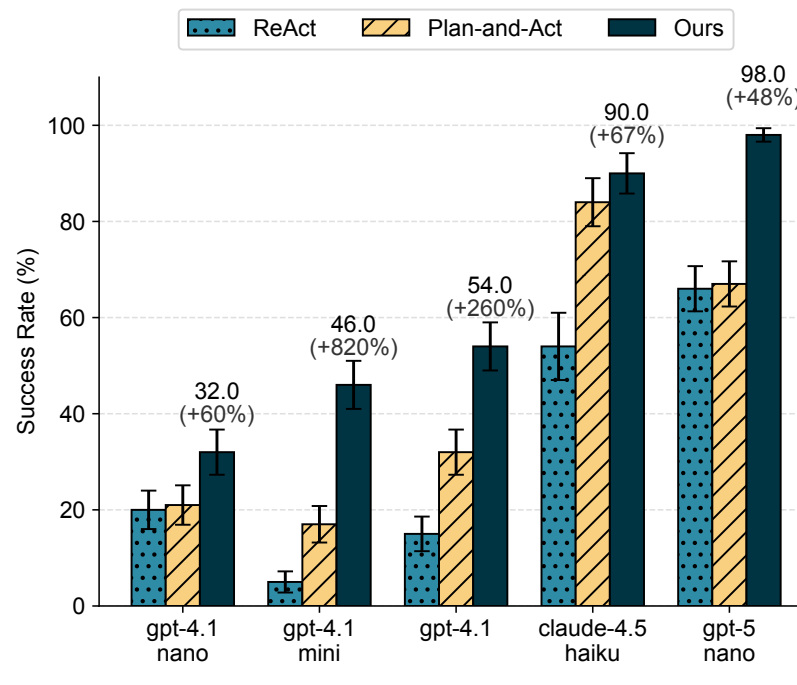


- TAPE shows better success rate on various tasks.



External Solver compensates weak LM planning ability

- +820% on weak gpt-4.1-mini
- +48% even on the strongest gpt-5-nano.



Error Analysis

- External solver mitigates planning errors.
- Constrained decoding drives sampling error to ~0%

Framework	Planning Error (%)	Sampling Error (%)	Success Rate (%)
ReAct	50.7	8.3	5.0
Plan-and-Act	47.7	4.7	17.0
TAPE (Ours)	36.7	0.0	46.0

Beyond ReAct: Works with Tree-of-Thought

- With Tree-of-Thought (ToT) as the candidate generator: ToT (10%) → ToT + TAPE (**25%**) on Sokoban.

Method	Solver	Constrained Exec.	Replan	Success Rate (%)
ToT	—	—	—	10.0
ToT + TAPE	✓	✓	✓	25.0

Code

Paper

Interesting?

